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A 36.34-day Sub-Neptune Candidate Transiting the Nearby K Dwarf TIC 402710280

Rishit Vora[1]

[1] *Independent Researcher, Bengaluru, India*

ABSTRACT

Transiting sub-Neptunes on orbits longer than a month are rare around bright stars. We report a candidate with a 36.34-day period around TIC 402710280, a $V = 10.0$ K dwarf at 66.8 pc that the Transiting Exoplanet Survey Satellite (*TESS*) has observed only in full-frame images. Four transits from 2021 to 2026, reduced by three independent pipelines, have consistent depths (mean 2092 ± 47 ppm) and a duration of 5.27 ± 0.05 h. Two are 36.36 days apart. The light curves exclude every shorter period consistent with both down to one day, and the duration disfavors still shorter ones, leaving $P = 36.33986 \pm 0.00051$ d. The candidate's radius is $3.75 \pm 0.22\,R_\oplus$ and its equilibrium temperature 498 ± 21 K. Difference-image centroids place the signal $1.5'' \pm 1.3''$ from the target, and eclipse models of the only nearby *Gaia* source, $3.5''$ away, fit worse than a planet by $\Delta\chi^2 \geq 142$ for every main-sequence pair. `TRICERATOPS` gives a false-positive probability of 0.003 ± 0.004; without high-resolution imaging, the object remains a candidate. The transit times scatter about a linear ephemeris by 21.8 min (rms), and an independent reduction reproduces the largest departure. Four transits cannot separate timing systematics from transit-timing variations. A second signal, of 407 ± 26 ppm at 3.78 days, may come from the target, an unresolved companion, or the background star. We predict a transit in *TESS* Sector 114 on 2027 February 19.

Keywords: Exoplanets (498) — Transit photometry (1709) — Mini Neptunes (1063) — K dwarf stars (876)

1. INTRODUCTION

The NASA Exoplanet Archive lists six confirmed transiting planets of 3–4 $R_\oplus$ around K dwarfs ($T_{\rm eff} = 3900$–5300 K) brighter than $V = 10.5$, and only one of them, TOI-2076 d ($P = 35.1$ d; C. Hedges et al. 2021), has a period longer than 30 days.[2] Such planets are warm rather than strongly irradiated, and around stars this bright their masses are measurable with high-precision radial velocities. A single sector of the Transiting Exoplanet Survey Satellite (*TESS*; G. R. Ricker et al. 2015), about 27 d long, records at most one transit of such a planet, so finding them requires combining sectors observed years apart.

Stars observed only in full-frame images lie outside the two-minute transit search of the Science Processing Operations Center (SPOC; J. M. Jenkins et al. 2016). Their long-period planets must be found in full-frame light curves, such as those of the Quick-Look Pipeline (QLP; C. X. Huang et al. 2020a,b) and TESS-SPOC (D. A. Caldwell et al. 2020a), which reach the most recent images only weeks to months after they are taken. TIC 402710280 shows the difficulty. Until 2026, its public light curves held two transits four years apart, and the only intervening epoch that *TESS* observed fell on a scattered-light ramp, so the period could not be measured from public data.

Here we report a sub-Neptune candidate with a period of 36.34 days around TIC 402710280, a K dwarf at 67 pc. We found its first two transits in public full-frame light curves and fixed the period with our own photometry of the 2026 full-frame images. As of 2026 October 18, the star is neither a *TESS* Object of Interest (TOI; N. M. Guerrero et al. 2021) nor a community TOI, SPOC has reported no threshold-crossing event for it, and neither the Exoplanet Follow-up Observing Program (ExoFOP), the NASA Exoplanet Archive (R. L. Akeson et al. 2013), nor the NASA Astrophysics Data System lists planet parameters, follow-up observations, or publications for it. We call the candidate

Email: rishitv@protonmail.com

[2] Planetary Systems Composite Parameters table (NASA Exoplanet Science Institute 2020), queried on 2026 October 21 for transiting planets with $3 \leq R_p/R_\oplus \leq 4$, $V < 10.5$, and $3900 \leq T_{\rm eff}/{\rm K} \leq 5300$; the query and the resulting table are in the data record (R. Vora 2026a).

TIC 402710280.01, the designation ExoFOP gives the first community TOI on a star, and will submit it as a community TOI once this paper is published in a refereed journal, as ExoFOP requires.

2. HOST STAR

Table 1 lists the properties of TIC 402710280. We adopt the stellar parameters of the *TESS* Input Catalog v8.2 (TIC; K. G. Stassun et al. 2019). The *Gaia* DR3 (Gaia Collaboration et al. 2016, 2023) astrophysical parameters agree with them: the GSP-Phot effective temperature is 5089 K (R. Andrae et al. 2023), and the FLAME radius and mass, $0.843 \pm 0.018\,R_\odot$ and $0.81 \pm 0.04\,M_\odot$ (O. L. Creevey et al. 2023), lie within 1σ of the TIC values. The FLAME radius would make the candidate 3% larger.

Gaia DR3 reports a renormalized unit weight error (RUWE; L. Lindegren et al. 2021) of 1.00, no multi-peaked image windows (`ipd_frac_multi_peak` = 0), and no non-single-star solution, as expected for a single star. These indicators do not exclude a faint companion at small separation.

The light curves show no rotational modulation. The strongest Lomb–Scargle peaks of individual sectors lie between 5 and 13 d, differ from sector to sector, and stay below 0.1% peak to peak in the TESS-SPOC light curves, whose systematics correction suppresses longer periods. We find no flares: no light curve has two or more consecutive cadences more than 4σ above its running median, except during one scattered-light episode in Sector 93 (BTJD 3830.6), when the excess follows the background. There is therefore no stellar signal against which to phase the transits.

3. OBSERVATIONS AND PHOTOMETRY

TESS observed TIC 402710280 only in full-frame images, in Sectors 12 (2019), 39 (2021), 66 (2023), 93 (2025), and 100–103 (2026), at cadences of 1800 s in the prime mission, 600 s in the first extended mission, and 200 s thereafter. We use the TESS-SPOC light curves (pre-search data conditioning flux; D. A. Caldwell et al. 2020a,b) of Sectors 12, 39, and 66 and the QLP light curve (systematics-corrected flux; C. X. Huang et al. 2020a,c) of Sector 93. For Sectors 100–103 we extracted light curves from 21×21-pixel cutouts (C. E. Brasseur et al. 2019) of the SPOC-calibrated full-frame images (TESS Team 2022). The background of each cadence is the median of the faintest 40% of pixels more than 4 pixels from the target. The aperture is a circle whose radius, chosen between 1 and 4 pixels in steps of 0.5, minimizes the one-hour scatter of the detrended light curve; the adopted radii are 1.5–2.5 pixels. We removed cadences flagged by SPOC and scattered-light glints, in which the background exceeds its one-day running median by more than eight times its median absolute deviation. QLP light curves of Sectors 100–103 were released at MAST between 2026 March and July but were absent from the public cloud copy of the archive that our search used. We use them only to check our transit times and depths (§5.3).

The time stamps of the calibrated full-frame images are barycentric for a reference position on the detector, not for the target. Recomputing them for the target's position changes them by −48 s and −20 s at the Sector 100 and 102 transits. The TESS-SPOC and QLP light curves we fit already refer to the target: their time stamps precede those of the same cadences in the calibrated images by 20.7 s, in agreement with the 20–23 s we compute.

The TESS-SPOC fluxes are corrected for light from neighboring stars (`CROWDSAP` = 0.92–0.94); our photometry is not. The depths from the two kinds of light curve agree within their 4–7% uncertainties (§5.1), so any difference in dilution between them is smaller than the measurement precision.

For the transit analysis, we extract each transit in a window of ±0.8 d and normalize it by a quadratic function of time fitted to the out-of-transit data.

4. DETECTION AND PERIOD

We found the candidate in a search of 10,324,827 public light curves for individual transit-like events: QLP light curves of Sectors 1–98 and SPOC two-minute light curves (TESS Team 2021) of Sectors 99–107. We applied box filters of ten durations between 0.75 and 18 h and kept events above 6σ relative to the scatter of each filtered series. After we removed systematics shared by many stars, events at data edges and gaps, and non-planetary shapes, 2263 stars retained single transit-like events. We vetted the 142 of them brighter than $T = 10.5$ individually, at the pixel level and against catalogs of known objects, and searched the 27 that survived without a known period for further transits in our photometry of Sectors 99–108.

The public light curves of TIC 402710280 contain two dips of matching depth and duration, in Sectors 39 and 93, 1453.58 d apart. Our photometry revealed two more transits, in Sectors 100 and 102, 36.36 d apart (Figure 1). Two transits 36.36 d apart allow only periods of $36.36/k$ d for integer k. We tested every k from 2 to 36, down to 1.0 d, by measuring the depth at every transit time each period implies within the data, excluding the four observed transits. The exclusion significances below include the uncertainty of the observed

Table 1. Properties of TIC 402710280

Parameter	Value	Source
R.A. (J2000)	16:34:05.67	TIC
Decl. (J2000)	−63:32:58.8	TIC
Gaia DR3 ID	5828441289812582400	*Gaia*
$\mu_\alpha \cos\delta$ (mas yr^{-1})	59.909	*Gaia*
μ_δ (mas yr^{-1})	13.685	*Gaia*
Parallax (mas)	14.964 ± 0.014	*Gaia*
Distance (pc)	66.83 ± 0.06	*Gaia*
RUWE	1.00	*Gaia*
T (mag)	9.243 ± 0.006	TIC
G (mag)	9.781	*Gaia*
$G_{\rm BP} - G_{\rm RP}$ (mag)	1.023	*Gaia*
V (mag)	9.982 ± 0.004	TIC
J (mag)	8.528 ± 0.026	TIC
K_s (mag)	8.026	TIC
Contamination ratio	0.060	TIC
$T_{\rm eff}$ (K)	5182 ± 121	TIC
$\log g$ (cgs)	4.56 ± 0.08	TIC
$R_\star$ ($R_\odot$)	0.817 ± 0.047	TIC
$M_\star$ ($M_\odot$)	0.88 ± 0.11	TIC
$\rho_\star$ ($\rho_\odot$)	1.61 ± 0.38	derived
$L_\star$ ($L_\odot$)	0.430 ± 0.005	*Gaia* FLAME

NOTE—$\rho_\star$ is computed from the TIC mass and radius. The contamination ratio is the TIC estimate of the flux from other stars relative to the target's within a *TESS* aperture. J and K_s are 2MASS magnitudes.

depth. They use white-noise uncertainties for the windows, and correlated noise as large as that of the transit windows ($\beta \leq 1.5$, Table 2) would reduce them by up to a third. At $P/2$, transits are expected in three covered windows (Sectors 39, 101, and 102) and are absent from each at $\geq 23.8\sigma$; the other $P/2$ epochs in the observed sectors fall in the mid-sector gap of Sector 12, before the first good cadence of Sector 100, or after the end of the Sector 93 data. At $P/3$, transits are absent from eight windows at $\geq 23.5\sigma$, and at $P/4$ and $P/5$ from 27 more at $\geq 17.1\sigma$. Each shorter period, from $P/6$ to $P/36$, places 19 to 152 transits in the data, and their mean depth falls short of the observed one by at least 41σ. Periods below 1.0 d are strongly disfavored by the duration. The longest transit such an orbit allows, centered at apastron on an eccentric orbit whose periastron touches the stellar surface, lasts 4.8 h around a star of the cataloged density, compared with the observed 5.27 ± 0.05 h, and reaches the observed duration only for a density 1.9σ below the cataloged value. At the adopted period, the Sector 12 data (BTJD 1624.9–1652.9, where BTJD = $\mathrm{BJD_{TDB}} - 2457000$) fall between transits at BTJD 1621.5 and 1657.8, Sector 101 contains no transit, and the Sector 103 transit, at BTJD 4165.26, falls in the data gap between BTJD 4164.44 and 4166.79. A period of 36.34 d is therefore the only one consistent with every covered epoch.

The transit expected in Sector 66 (BTJD 3111.40) coincides with strong scattered light (Figure 2). The background falls from 1.5×10^4 to 2.9×10^3 e$^-$ s^{-1} across the window, the first cadences without quality flags begin 3.0 h before the predicted mid-time (0.4 h before ingress), and QLP flags every cadence within 12 h of it. The transit is covered, but the baseline before it is not. Fitted with the shape of §5.2 and a free mid-time, the event has depths of 1380, 1270, and 1620 ppm (uncertainties 100–160 ppm) for constant, linear, and quadratic baselines and lies −5 to +2 min from the linear ephemeris of §5.3. Its timing is robust to the base-

line, but its depth is not, so we exclude Sector 66 from all fits of the transits and from the ephemeris.

5. TRANSIT ANALYSIS

5.1. *Individual Transit Times and Depths*

We timed each transit by fixing the shape to the median model of §5.2 and scanning the mid-time on a 0.5-minute grid over ±2 h, refitting a quadratic baseline across the ±0.8 d window at each step. The 1σ uncertainty is the half-width of the $\Delta\chi^2 = 1$ interval multiplied by the red-noise factor β (F. Pont et al. 2006; J. N. Winn et al. 2008), the median ratio of the binned residual scatter (10–60-minute bins) to its white-noise expectation. We measured each depth with the shape fixed and the amplitude free. Table 2 lists the results.

The four depths agree ($\chi^2 = 2.5$ for 3 degrees of freedom), although they come from three independent reductions with different apertures, cadences, and crowding corrections. This agreement argues against an origin in any one pipeline and against aperture-dependent dilution. Their weighted mean is 2092 ± 47 ppm.

5.2. *Transit Shape*

We fitted the four transits jointly with `batman` (L. Kreidberg 2015), integrating over each light curve's exposure time and centering each window on its measured mid-time so that the timing scatter (§5.3) does not smear the stacked profile. The free parameters are $R_p/R_\star$, b, and $\log_{10}\rho_\star$, with uniform priors; $a/R_\star$ follows from $\rho_\star$ and P for a circular orbit (S. Seager & G. Mallén-Ornelas 2003). The quadratic limb-darkening coefficients are fixed at $(u_1, u_2) = (0.39, 0.21)$, interpolated in $T_{\rm eff}$ from the *TESS*-band tables of A. Claret (2017) at $\log g = 4.5$. Each window's flux uncertainty is its out-of-transit point-to-point scatter multiplied by its β. We sampled the posterior with `emcee` (J. Goodman & J. Weare 2010; D. Foreman-Mackey et al. 2013), running 40 walkers for 30,000 steps and discarding the first third; the chain spans more than 90 integrated autocorrelation times for every parameter. Table 3 gives the medians and 68% intervals. The best-fitting model has a reduced χ^2 of 0.84 for 2227 data points, so the β-scaled uncertainties are slightly conservative.

The impact parameter is below 0.54 at 95% confidence. The fitted stellar density, $1.57^{+0.12}_{-0.31}\,\rho_\odot$, agrees with the $1.61 \pm 0.38\,\rho_\odot$ derived from the TIC mass and radius, as expected for a circular orbit around a star of the cataloged density. Fitting the limb darkening as well, with the uninformative parametrization of D. M. Kipping (2013), changes $R_p/R_\star$ to 0.0426 ± 0.0006 and b to $0.21^{+0.19}_{-0.15}$, both within 1σ of the adopted values.

5.3. *Ephemeris and Timing Residuals*

A weighted linear fit to the four mid-times gives $P = 36.33986 \pm 0.00051$ d and $T_0 = 2460838.2008 \pm 0.0119$ $\mathrm{BJD_{TDB}}$, with the reference epoch at the Sector 93 transit (epoch 40). The weighted mean epoch is 29, so T_0 and P are correlated (correlation coefficient +0.47), and the predictions of §9 include their covariance. The residuals (Table 2, Figure 3) are +4.4, −23.2, +6.8, and +35.9 min. Their rms of 21.8 min far exceeds the formal uncertainties of 1.7–3.3 min ($\chi^2 = 417$ for 2 degrees of freedom), and we scale the ephemeris uncertainties by $\sqrt{\chi^2/\nu}$. The largest single contribution is the interval between the consecutive Sector 100 and 102 transits, which exceeds P by 29.2 min.

We looked for an instrumental cause. Changing the baseline (constant, linear, or quadratic) or the window half-width (0.6, 0.8, or 1.0 d) moves no mid-time by more than 2.3 min. The Sector 93 mid-time agrees within 1.9 min across every QLP flux series: the systematics-corrected flux, the raw flux, and the detrended fluxes of the optimal, small, and large apertures. That window is complete (688 of 691 cadences, no gap longer than 7 min) and shows no background or centroid feature at the transit. The time stamps of all four light curves refer to the target's barycentric position (§3). Masking the transits of the 3.78-day signal (§7), one of which falls 4 h before the Sector 93 transit, moves no mid-time by more than 0.1 min.

The QLP light curves of Sectors 100 and 102 provide an independent reduction of the same images, but their time stamps must first be corrected. Matched to the full-frame images by cadence number, they precede the mid-exposure times of the images, corrected to the target, by 121 and 139 s at the two transits; in Sector 93 the two agree within 3 s. The offset changes by −6.3 and −6.1 s per day, the negatives of the rates at which the target's barycentric correction changes (6.3 and 6.1 s per day), and the QLP times of Sectors 100 and 102 are linear in cadence number to within 0.01 s across each sector. Both results point to a single barycentric correction applied to every cadence of each sector. The offset vanishes, and the applied correction equals the target's, near BTJD 4072, two days before the first Sector 100 cadence (BTJD 4074.0), and near BTJD 4107, during Sector 101 and three weeks before the first Sector 102 cadence (BTJD 4127.0). QLP adopted the photometry of T. Han & T. D. Brandt (2023) from Sector 94 onward (G. Petitpas et al. 2026); the Sector 93 time stamps, which follow the barycentric correction, precede that change. We have reported the offset to the QLP team.

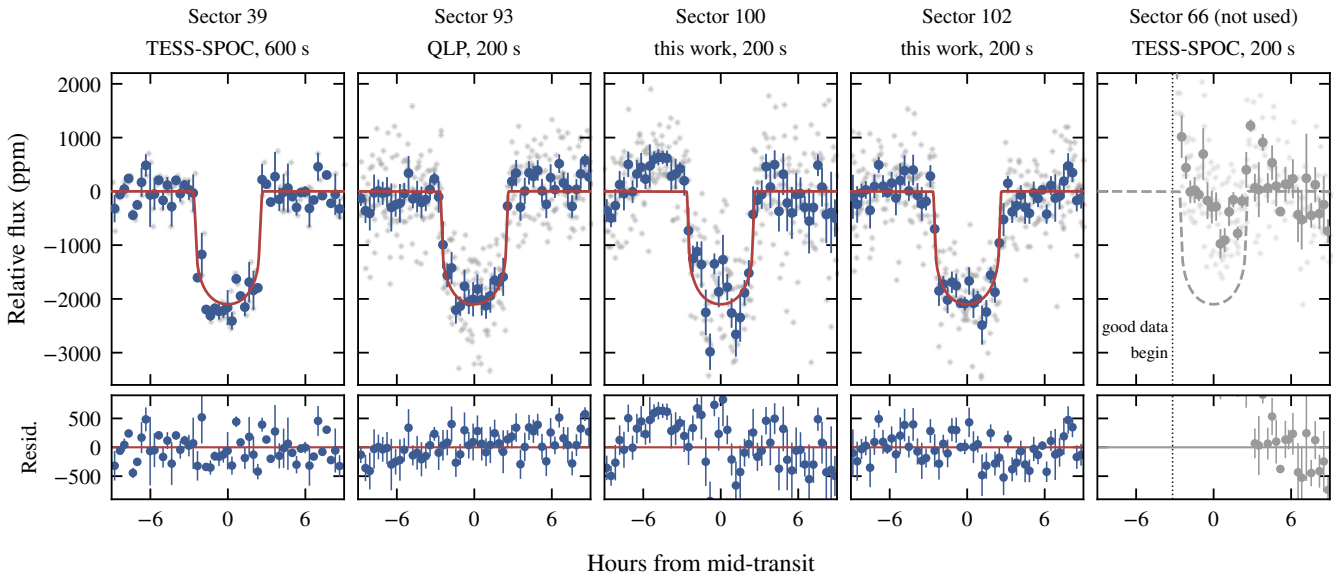


Figure 1. Transits of TIC 402710280.01. Gray points are individual cadences, and blue points are 20-minute bins with error bars equal to the standard error of the mean. Each window is centered on its measured mid-time (Table 2), and the red curve is the median transit model of §5.2; the lower panels show the binned residuals. Our Sector 100 and 102 photometry has a larger per-cadence scatter than the QLP light curve at the same cadence (Table 2) because our apertures are smaller and are not optimized per cadence. The right-hand panel shows the Sector 66 window at the time predicted by the linear ephemeris; cadences without quality flags begin at the dotted line, and the window enters no fit (§4).

Table 2. Individual transits of TIC 402710280.01

Sector	Light curve	Cadence (s)	σ (ppm)	β	Epoch	T_c ($\mathrm{BJD_{TDB}}$)	$O-C$ (min)	Depth (ppm)
39	TESS-SPOC	600	355	1.00	0	2459384.6094 ± 0.0013	+4.4	2203 ± 86
93	QLP	200	594	1.02	40	2460838.1847 ± 0.0012	−23.2	2025 ± 84
100	This work	200	702	1.47	47	2461092.5845 ± 0.0023	+6.8	2078 ± 144
102	This work	200	608	1.08	48	2461128.9446 ± 0.0017	+35.9	2053 ± 91

NOTE—σ is the out-of-transit scatter per cadence and β the red-noise factor of each window; every window spans ±0.8 d and has a quadratic baseline. $O-C$ is relative to the linear ephemeris of Table 3. Depths are the model depth at mid-transit for a free amplitude with the shape fixed.

With this correction, and timed as in §5.1, the QLP transits fall 4.1 and 4.6 min earlier than ours (0.7 and 1.5 times the combined uncertainty), and the interval between them exceeds P by 28.7 min, compared with 29.2 min in our photometry. Our photometry of the Sector 39 and 93 transits, reduced in the same way, gives mid-times within 0.8 min of the TESS-SPOC and QLP values. The excess in the Sector 100–102 interval therefore does not arise in our reductions.

The QLP depths of the Sector 100 and 102 transits, 1757 ± 170 and 1666 ± 83 ppm, are lower than ours by 1.4 and 3.1 times the combined uncertainty, whereas the Sector 93 QLP depth, measured before the change of method, is 2025 ± 84 ppm and agrees with the others. Our photometry is not corrected for neighboring stars, which would make its depths too shallow rather than too deep, so the difference does not come from our treatment of crowding. We have not identified its cause, and neither QLP light curve enters our fits. Adopting the QLP depths of these two transits would lower the mean depth to 1943 ± 47 ppm and the radius to $3.61\,R_\oplus$, but the four depths would then disagree ($\chi^2 = 22$ for 3 degrees of freedom). Adopting the QLP mid-times would change P by 4 s and the timing rms to 19.3 min.

We identify no instrumental cause of the timing scatter in the light curves we fit. Four epochs cannot distinguish timing systematics from transit-timing variations induced by another body, and further transits will show

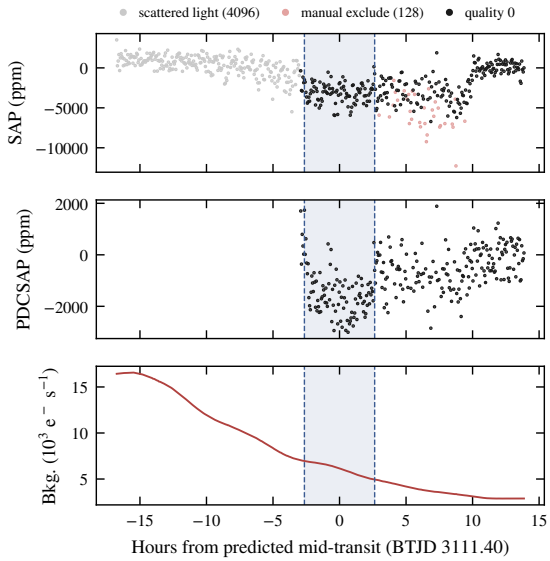


Figure 2. TESS-SPOC data of Sector 66 around the predicted transit, whose duration is shaded and whose ingress and egress are marked by dashed lines. From top: simple-aperture flux, with cadences flagged by SPOC for scattered light (quality value 4096) in gray and as manually excluded (value 128) in pink; pre-search data conditioning flux, quality 0 only; and the background.

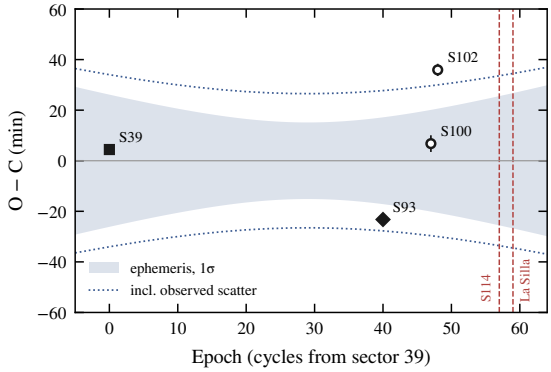


Figure 3. Transit-timing residuals relative to the linear ephemeris. Symbols denote the light-curve source: TESS-SPOC (square), QLP (diamond), and our photometry (open circles). The formal uncertainties, 1.7–3.3 min, are smaller than the symbols. The shaded band is the 1σ uncertainty of the ephemeris, and the dotted curves add the 21.8-min rms scatter of the residuals in quadrature. Dashed lines mark epochs 57 (2027 February 19, *TESS* Sector 114) and 59 (2027 May 3, observable from La Silla).

whether the residuals are periodic. The predictions of §9 include the observed scatter.

6. VETTING

Without high-resolution imaging or spectroscopy, the light curves and the *Gaia* catalog test four things: whether the signal comes from the target's position (§6.1), whether the one cataloged star close enough to contribute can produce it (§6.2), whether the target is itself an eclipsing binary (§6.3), and how probable the remaining false-positive scenarios are, given the field (§6.4). They cannot exclude an unresolved companion within about 1″.

Table 3. Parameters of TIC 402710280.01

Parameter	Value
Ephemeris	
P (d)	36.33986 ± 0.00051
T_0 ($\mathrm{BJD_{TDB}}$)	2460838.2008 ± 0.0119
Timing residual rms (min)	21.8
Fitted	
$R_p/R_\star$	$0.04209^{+0.00067}_{-0.00051}$
b	$0.22^{+0.20}_{-0.15}$
$\rho_\star$ ($\rho_\odot$)	$1.57^{+0.12}_{-0.31}$
Fixed	
e	0
(u_1, u_2)	(0.39, 0.21)
Derived	
Depth at mid-transit (ppm)	2092 ± 47
T_{14} (h)	5.27 ± 0.05
$a/R_\star$	$53.6^{+1.3}_{-3.7}$
R_p ($R_\oplus$)	3.75 ± 0.22
a (AU)	0.206 ± 0.009
S ($S_\oplus$)	10.2 ± 1.7
$T_{\rm eq}$ (K)	498 ± 21

NOTE—Medians with 68% intervals. The depth is the weighted mean of the four measurements of Table 2. $T_{\rm eq}$ assumes zero Bond albedo and full heat redistribution. R_p, a, S, and $T_{\rm eq}$ use the stellar parameters of Table 1.

6.1. *Location of the Signal*

For each transit we formed a difference image (out of transit minus in transit) from the full-frame cutouts and fitted a Gaussian point-spread function to it (cf. S. T. Bryson et al. 2013; J. D. Twicken et al. 2018). We measured each offset relative to the target's own fitted position in the out-of-transit image, which removes errors in the world-coordinate solution and the effect of proper motion. The four transits combined place the signal $1.5'' \pm 1.3''$ from TIC 402710280 (Figure 4). Applied to 40 confirmed *TESS* planets, the method placed 30 on their host star, left 5 unresolved from a nearby source, returned an offset without a cataloged counterpart for 3, failed for 2, and placed none on a neighboring star.

The centroid lies 1.2σ from the target and 1.6σ from one neighbor. Every other *Gaia* DR3 source bright enough to produce the observed depth lies 21σ or more away and is excluded; this one is not. The neighbor, *Gaia* DR3 5828441289812582656 ($G = 16.15$, $\Delta G = 6.37$), has a parallax of 0.69 ± 0.08 mas and is therefore in the background. Because of the target's proper motion, the separation grows from $3.10''$ at the *Gaia* epoch (2016.0) to $3.4''$–$3.6''$ during the transits.

6.2. *Blended Eclipsing Binary*

Diluted by TIC 402710280, the neighbor could produce the observed depth only by losing about 74% of its light in eclipse. We fitted the four transits with eclipse models of the neighbor for companion-to-star radius ratios k from 0.3 to 10, with the impact parameter and $a/R_\star$ free, the dilution free but bounded at 0.4% of the total flux (40% above the star's share, to allow for its unknown *TESS*-band color), and three limb-darkening laws (Figure 5). For $k \leq 1.2$, every eclipse model fits worse than the planet model by $\Delta\chi^2 \geq 142$ for 2227 data points. Larger companions fit better ($\Delta\chi^2 = 113$, 70, 44, 30, and 18 for $k = 1.5$, 2, 3, 5, and 10), because a total eclipse with a short ingress resembles a transit, but only if the eclipsed star supplies about 70% of the neighbor's light and a companion 1.5 to 10 times its radius supplies the rest. The companion would then need a surface brightness 5 ($k = 1.5$) to 200 ($k = 10$) times lower than the eclipsed star's, which no main-sequence pair has; only a hot, compact eclipsed star could produce such an eclipse. A blend with the neighbor is therefore disfavored, and a ground-based light curve with an aperture that excludes the neighbor would settle the question.

6.3. *Secondary Eclipse and Odd–Even Depths*

The light curve shows no eclipse at orbital phase 0.5, which Sectors 39, 101, and 102 sample: the weighted depth there is -39 ± 39 ppm, below 77 ppm at 3σ. A search over phases 0.3 to 0.7, which covers secondary eclipses of eccentric orbits with $|e\cos\omega| \lesssim 0.3$, finds no consistent signal. The 3σ upper limit is below 160 ppm at half of these phases and below 365 ppm at all of them. The most significant box, at phase 0.449 (204 ± 53 ppm), is dominated by data from the last 0.5 d of Sector 93 and is not reproduced in Sectors 39 and 101. These limits exclude an eclipsing companion with a surface brightness comparable to the target's. Only one transit (Sector 100) falls on an odd epoch, so odd and even depths cannot be compared usefully, but the agreement of all four depths (§5.1) disfavors an eclipsing binary at twice the period with unequal eclipses.

6.4. *Statistical False-Positive Probability*

Statistical validation with `TRICERATOPS` (S. Giacalone & C. D. Dressing 2020; S. Giacalone et al. 2021) requires high-resolution imaging, and none exists for this star. Without a contrast curve, unresolved companions are constrained only by population priors, so the probabilities below do not validate the candidate. We computed scenario probabilities from the four transits stacked on their measured mid-times (10-minute bins), a depth of 2100 ppm, the photometric apertures of Sectors 39, 100, and 102, a circle of 2 pixels radius in Sector 93 (the QLP aperture has a radius of 3 pixels), and the *Gaia* DR3 field. Ten independent runs give a false-positive probability FPP $= 0.003 \pm 0.004$ (range 0.000–0.012) and a nearby false-positive probability NFPP $< 10^{-6}$.[3] `TRICERATOPS` places the stars at their TIC positions, at epoch J2000, where the neighbor lies $2.3''$ from the target rather than the $3.4''$–$3.6''$ of the transits (§6.1). The smaller separation puts more of the neighbor's light in the aperture, so the computed NFPP overstates the nearby false-positive probability. A planet on the target receives 85% of the probability and a planet diluted by an unresolved bound companion 14%; the leading false-positive scenario, a planet orbiting an unresolved companion, receives 0.3%. These values meet the numerical criteria of S. Giacalone et al. (2021), FPP < 0.015 and NFPP < 0.001, but without imaging we classify TIC 402710280.01 as a planet candidate.

7. A SECOND PERIODIC SIGNAL

With the transits of TIC 402710280.01 masked, a box-least-squares search (G. Kovács et al. 2002) of the light curve finds a second periodic signal, with $P = 3.78371$ d, a depth of 407 ± 26 ppm, and a duration of about 1.3 h (Figure 6). The search detects it independently in Sectors 12–93, as the strongest periodicity (signal detection efficiency 14.4), and in Sectors 100–103, as the second strongest after its 5:1 alias at 18.92 d. The two ephemerides agree in phase across 650 cycles. The depths in the eight sectors agree with one another ($\chi^2 = 7.3$ for 7 degrees of freedom), the odd and even transits agree (382 ± 39 and 437 ± 46 ppm), and no eclipse appears at phase 0.5 (-61 ± 26 ppm).

A transit model with free stellar density gives $R_p/R_\star = 0.0196^{+0.0015}_{-0.0008}$, $b = 0.49 \pm 0.32$, and $\rho_\star = 8.3^{+3.7}_{-5.5}\,\rho_\odot$. On the target, the signal would correspond to a planet of $1.75 \pm 0.11\,R_\oplus$ on an inner orbit. The fitted density, $8.3^{+3.7}_{-5.5}\,\rho_\odot$, exceeds the target's $1.61 \pm 0.38\,\rho_\odot$.

[3] With the photometric apertures, NFPP is numerically zero ($< 10^{-20}$); with the default 5×5-pixel apertures of `TRICERATOPS`, FPP $= 0.001$ and NFPP $= 6 \times 10^{-7}$.

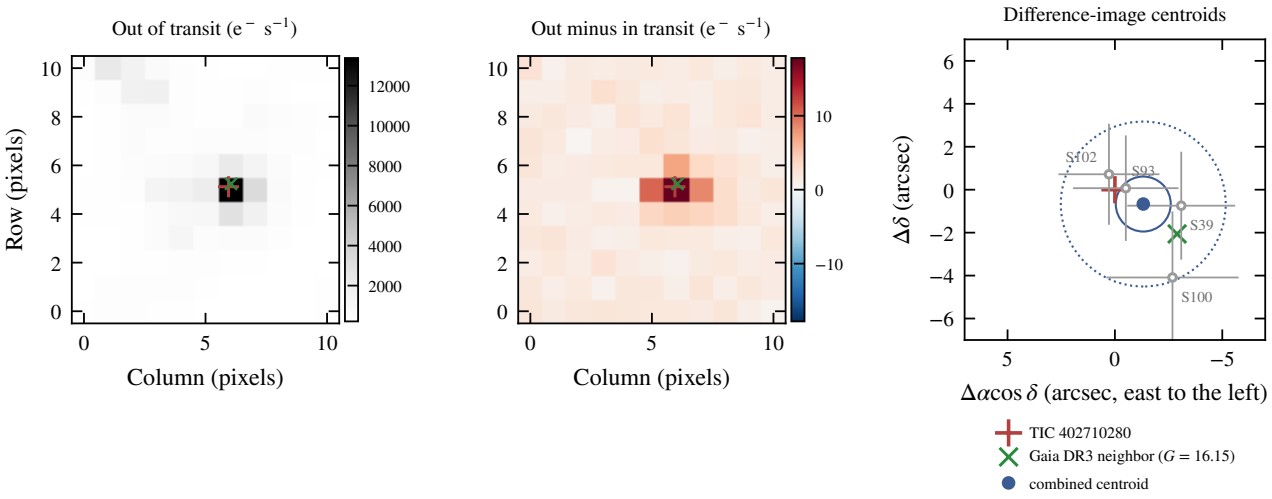


Figure 4. Location of the signal. Left and center: the out-of-transit image and the difference image (out of transit minus in transit) of the Sector 39 transit, which has the highest signal-to-noise ratio. The plus marks TIC 402710280 and the cross the background neighbor; the two are not separated at the 21″ pixel scale. Right: the centroid offsets of the four transits (gray) and their combination (blue), with 1σ (solid) and 3σ (dotted) circles, relative to TIC 402710280. The cross marks the neighbor at the mean epoch of the transits.

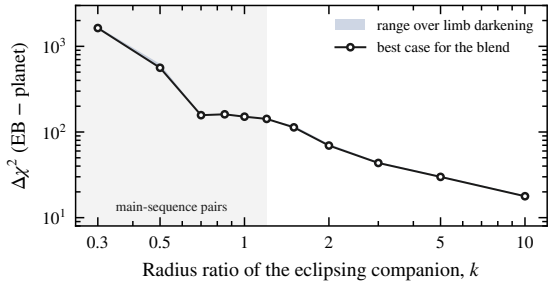


Figure 5. Diluted-eclipse models for the background neighbor, compared with the planet model. For each companion radius ratio k, the curve gives the smallest χ^2 difference over impact parameter, $a/R_\star$, dilution, and limb darkening, and the band gives the range over the three limb-darkening laws. The shaded region, $k \leq 1.2$, contains every main-sequence pair.

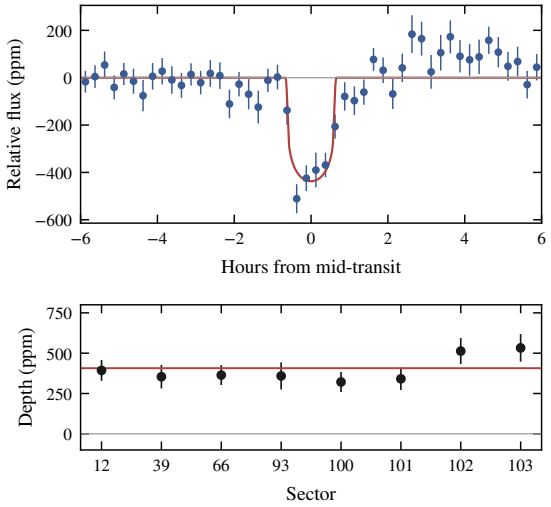


Figure 6. The 3.78-day signal. Top: all eight sectors folded on its ephemeris, with the transits of TIC 402710280.01 masked, in 15-minute bins (error bars are standard errors of the mean), and the median transit model. Bottom: the box depth in each sector, with uncertainties from boxes placed at 400 random phases; the line is the mean over all sectors.

Only 13% of the posterior lies below $2.37\,\rho_\odot$ (the target's value plus 2σ), almost all of it at $b > 0.82$, so a transit of the target matches the short duration only if it is nearly grazing. The difference-image centroid of this weaker signal, $5.2'' \pm 6.7''$ from the target, cannot separate the target from the background neighbor, which would have to lose about 14% of its light in eclipse to produce the signal. `TRICERATOPS`, applied as in §6.4 but with the 3-pixel QLP aperture in Sector 93, assigns 98% of the probability to a planet around the target or an unresolved companion but gives FPP $= 0.12 \pm 0.01$ and NFPP $= 0.015 \pm 0.001$, above the NFPP limit of 0.001 below which S. Giacalone et al. (2021) consider nearby false positives excluded. We therefore do not claim this signal as a planet candidate: its source may be the target, an unresolved companion, or the neighbor. A ground-based light curve that resolves the neighbor would show the 14% eclipse if the neighbor is the source.

8. DISCUSSION

With $R_p = 3.75 \pm 0.22\,R_\oplus$, TIC 402710280.01 lies above the radius valley (B. J. Fulton et al. 2017), among sub-Neptunes that probably retain a substantial volatile envelope. At 0.206 AU it receives about ten times

Earth's insolation. The mass–radius relation of J. Chen & D. Kipping (2017) predicts $M_p \approx 14\,M_\oplus$, which would induce a radial-velocity semi-amplitude of about 2.8 m s^{-1}, measurable with current high-precision spectrographs for a $V = 10.0$ star. For that mass, the transmission spectroscopy metric of E. M.-R. Kempton et al. (2018) is about 70, below the threshold of 90 that E. M.-R. Kempton et al. (2018) recommend for sub-Neptunes, so the planet is a moderate rather than a prime target for transmission spectroscopy. If confirmed, it would be the second planet of 3–4 $R_\oplus$ with a period longer than 30 days around a K dwarf brighter than $V = 10.5$, after TOI-2076 d.

Three observations would test the candidate most efficiently: high-resolution imaging, to exclude unresolved companions and complete the statistical validation; seeing-limited photometry of a transit with an aperture that excludes the 3.5″ neighbor, to locate both signals; and radial velocities, to measure the mass. Each could overturn the present interpretation. A companion within 1″ bright enough to dilute the transit would change the radius and the false-positive probability, a 74% eclipse of the neighbor at the time of a transit would identify the signal as a blend, and a semi-amplitude far above the few meters per second expected for a sub-Neptune would reveal a stellar or substellar companion.

9. PREDICTED TRANSITS

Table 4 lists three of the next transits predicted by the ephemeris of Table 3; the data record lists the next 30 (R. Vora 2026a). The quoted uncertainty, $\sigma_{\rm pred} = (\sigma_{T_0}^2 + n^2\sigma_P^2 + 2n\,{\rm Cov}(T_0,P) + \sigma_{\rm rms}^2)^{1/2}$ for n cycles from the reference epoch, adds the observed residual scatter, $\sigma_{\rm rms} = 21.8$ min, to the ephemeris uncertainty, so that it describes an individual transit rather than the mean ephemeris. *TESS* is expected to observe TIC 402710280 in Sectors 113 and 114 (C. J. Burke et al. 2020), and the transit of 2027 February 19 falls in Sector 114. The transit of 2027 May 3 is observable in full from La Silla. The test of these predictions, with the data to be used and the rules that decide its outcome, was registered before the data existed (R. Vora 2026b). The registered $\sigma_{\rm pred}$ omits the covariance term (30.4 instead of 33.6 min at epoch 57), so the registered search over $\pm 3\sigma_{\rm pred}$ spans $\pm 2.7\sigma$ of the uncertainty given here; we keep the registered test unchanged. Under the registered rules, a transit absent from its window in data that cover it falsifies the period. A timing departure larger than the half-width of that window, four times the observed rms, would have the same signature.

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The author used Claude (Anthropic), a large language model, throughout this work, under the author's direction and review: to write and debug the analysis code, to produce the figures from the analysis outputs, to search the literature, and to draft and edit the text. Every reference was checked against its DOI or ASCL record, and the analysis of the four transits can be reproduced from the archived code and data (R. Vora 2026a). Responsibility for the content lies entirely with the author.

Facilities: TESS, Gaia

Software: astropy (Astropy Collaboration et al. 2022), astrocut (C. E. Brasseur et al. 2019), batman (L. Kreidberg 2015), emcee (D. Foreman-Mackey et al. 2013), lightkurve (Lightkurve Collaboration 2018), matplotlib (J. D. Hunter 2007), numpy (C. R. Harris et al. 2020), scipy (P. Virtanen et al. 2020), tess-point (C. J. Burke et al. 2020), TRICERATOPS (S. Giacalone & C. D. Dressing 2020)

DATA AVAILABILITY

The light curves of all transit windows, our photometry of Sectors 100–103, the measured transit times, the comparison with the QLP light curves of Sectors 100 and 102 and the test of their time stamps, the

Table 4. Predicted transits of TIC 402710280.01

Epoch	T_c (BJD_{TDB})	σ_{pred} (min)	UTC at La Silla	Note
57	2461455.9785	34	2027 Feb. 19, 11:29	*TESS* Sector 114
58	2461492.3183	34	2027 Mar. 27, 19:35	daytime at La Silla
59	2461528.6582	34	2027 May 3, 03:41	full transit at night at La Silla

NOTE—Epochs count cycles from the Sector 39 transit. UTC times are topocentric at La Silla Observatory: the light-travel time and TDB−UTC are removed.

alias and secondary-eclipse tests, the archive comparison sample, the posterior summaries and chains, the next 30 predicted transits, the analysis code, and the intermediate results from which the code fills in the manuscript are available on Zenodo (R. Vora 2026a, version 4.0; all versions: doi:10.5281/zenodo.23390273). The search of §4, which selected the star, belongs to a wider survey and is not part of the record. The pre-registered test of the 2027 transits is a separate Zenodo record (R. Vora 2026b). All the *TESS* data used in this paper can be found in MAST: the TESS-SPOC light curves of Sectors 12, 39, and 66, https://doi.org/10.17909/t9-wpz1-8s54; the QLP light curves of Sectors 1–98 (search) and Sectors 93, 100, and 102 (analysis), https://doi.org/10.17909/t9-r086-e880; the calibrated full-frame images of Sectors 39, 93, and 100–103, https://doi.org/10.17909/0cp4-2j79; and the SPOC two-minute light curves of Sectors 99–107 (search), https://doi.org/10.17909/t9-nmc8-f686. The stellar parameters are from the *TESS* Input Catalog, https://doi.org/10.17909/fwdt-2x66.

REFERENCES

Akeson, R. L., Chen, X., Ciardi, D., et al. 2013, PASP, 125, 989, doi: 10.1086/672273

Andrae, R., Fouesneau, M., Sordo, R., et al. 2023, A&A, 674, A27, doi: 10.1051/0004-6361/202243462

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Brasseur, C. E., Phillip, C., Fleming, S. W., Mullally, S. E., & White, R. L. 2019, Astrocut: Tools for creating cutouts of TESS images, Astrophysics Source Code Library, record ascl:1905.007

Bryson, S. T., Jenkins, J. M., Gilliland, R. L., et al. 2013, PASP, 125, 889, doi: 10.1086/671767

Burke, C. J., Levine, A., Fausnaugh, M., et al. 2020, TESS-Point: High precision TESS pointing tool, Astrophysics Source Code Library, record ascl:2003.001

Caldwell, D. A., Tenenbaum, P., Twicken, J. D., et al. 2020a, Research Notes of the AAS, 4, 201, doi: 10.3847/2515-5172/abc9b3

Caldwell, D. A., et al. 2020b, TESS Light Curves From Full Frame Images ("TESS-SPOC"), STScI/MAST doi: 10.17909/t9-wpz1-8s54

Chen, J., & Kipping, D. 2017, ApJ, 834, 17, doi: 10.3847/1538-4357/834/1/17

Claret, A. 2017, A&A, 600, A30, doi: 10.1051/0004-6361/201629705

Creevey, O. L., Sordo, R., Pailler, F., et al. 2023, A&A, 674, A26, doi: 10.1051/0004-6361/202243688

Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, PASP, 125, 306, doi: 10.1086/670067

Fulton, B. J., Petigura, E. A., Howard, A. W., et al. 2017, AJ, 154, 109, doi: 10.3847/1538-3881/aa80eb

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1, doi: 10.1051/0004-6361/201629272

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1, doi: 10.1051/0004-6361/202243940

Giacalone, S., & Dressing, C. D. 2020, triceratops: Candidate exoplanet rating tool, Astrophysics Source Code Library, record ascl:2002.004

Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, AJ, 161, 24, doi: 10.3847/1538-3881/abc6af

Goodman, J., & Weare, J. 2010, Communications in Applied Mathematics and Computational Science, 5, 65, doi: 10.2140/camcos.2010.5.65

Guerrero, N. M., Seager, S., Huang, C. X., et al. 2021, ApJS, 254, 39, doi: 10.3847/1538-4365/abefe1

Han, T., & Brandt, T. D. 2023, AJ, 165, 71, doi: 10.3847/1538-3881/acaaa7

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Hedges, C., Hughes, A., Zhou, G., et al. 2021, AJ, 162, 54, doi: 10.3847/1538-3881/ac06cd

Huang, C. X., Vanderburg, A., Pál, A., et al. 2020a, Research Notes of the AAS, 4, 204, doi: 10.3847/2515-5172/abca2e
Huang, C. X., Vanderburg, A., Pál, A., et al. 2020b, Research Notes of the AAS, 4, 206, doi: 10.3847/2515-5172/abca2d
Huang, C. X., et al. 2020c, TESS Lightcurves From The MIT Quick-Look Pipeline ("QLP"), STScI/MAST doi: 10.17909/t9-r086-e880
Hunter, J. D. 2007, Computing in Science and Engineering, 9, 90, doi: 10.1109/MCSE.2007.55
Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, in Proc. SPIE, Vol. 9913, Software and Cyberinfrastructure for Astronomy IV, 99133E, doi: 10.1117/12.2233418
Kempton, E. M.-R., Bean, J. L., Louie, D. R., et al. 2018, PASP, 130, 114401, doi: 10.1088/1538-3873/aadf6f
Kipping, D. M. 2013, MNRAS, 435, 2152, doi: 10.1093/mnras/stt1435
Kovács, G., Zucker, S., & Mazeh, T. 2002, A&A, 391, 369, doi: 10.1051/0004-6361:20020802
Kreidberg, L. 2015, PASP, 127, 1161, doi: 10.1086/683602
Lightkurve Collaboration. 2018, Lightkurve: Kepler and TESS time series analysis in Python, Astrophysics Source Code Library, record ascl:1812.013
Lindegren, L., Klioner, S. A., Hernández, J., et al. 2021, A&A, 649, A2, doi: 10.1051/0004-6361/202039709
NASA Exoplanet Science Institute. 2020, Planetary Systems Composite Table, IPAC doi: 10.26133/NEA13
Petitpas, G., Haviland, J., Han, T., et al. 2026, RNAAS, 10, 69, doi: 10.3847/2515-5172/ae5790
Pont, F., Zucker, S., & Queloz, D. 2006, MNRAS, 373, 231, doi: 10.1111/j.1365-2966.2006.11012.x
Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, Journal of Astronomical Telescopes, Instruments, and Systems, 1, 014003, doi: 10.1117/1.JATIS.1.1.014003
Seager, S., & Mallén-Ornelas, G. 2003, ApJ, 585, 1038, doi: 10.1086/346105
Stassun, K. G., Oelkers, R. J., Paegert, M., et al. 2019, AJ, 158, 138, doi: 10.3847/1538-3881/ab3467
TESS Team. 2021, TESS Light Curves - All Sectors, STScI/MAST doi: 10.17909/t9-nmc8-f686
TESS Team. 2022, TESS Calibrated Full Frame Images: All Sectors, STScI/MAST doi: 10.17909/0cp4-2j79
Twicken, J. D., Catanzarite, J. H., Clarke, B. D., et al. 2018, PASP, 130, 064502, doi: 10.1088/1538-3873/aab694
Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2
Vora, R. 2026a, TIC 402710280.01: data products and analysis code, Zenodo, version 4.0 doi: 10.5281/zenodo.23553448
Vora, R. 2026b, A Pre-registered Test of the Transiting Planet Candidate TIC 402710280.01 at Its Predicted Transits of 2027, Zenodo doi: 10.5281/zenodo.23390325
Winn, J. N., Holman, M. J., Torres, G., et al. 2008, ApJ, 683, 1076, doi: 10.1086/589737